

Yes. There are several **fundamental concepts** that you should know if you are building Company Brain. These concepts are more important than the code because they explain **why** the system is designed this way. I'll explain them in the order that data flows through your platform.

1. Documents are NOT your final data

Many people think:

```
graph TD; A[PDF] --> B[Store PDF]; B --> C[Ask AI];
```

PDF

↓

Store PDF

↓

Ask AI

That's how a basic RAG system works.

Company Brain works differently.

```
graph TD; A[PDF] --> B[Knowledge]; B --> C[Company Memory]; C --> D[Reasoning]; D --> E[Decision Intelligence];
```

PDF

↓

Knowledge

↓

Company Memory

↓

Reasoning

↓

Decision Intelligence

The document is only the **source**.

The real product is the **knowledge extracted from it**.

2. There are four different types of information

Most beginners mix these together.

Type 1 — Entity

An entity is a **thing**.

Examples:

Alice

Microsoft

Project Phoenix

Legal Approval

Customer ABC

Budget.xlsx

Everything that exists becomes an entity.

Type 2 — Relationship

A relationship connects two entities.

Example

Alice

↓

works_on

↓

Project Phoenix

Another

Project Phoenix

↓

depends_on

↓

Legal Approval

Relationships become graph edges.

Type 3 — Fact

Facts describe entities.

Example

Project Phoenix

Budget

\$2.8M

or

Project Phoenix

Status

Active

Notice

Budget is NOT an entity.

It is a property.

Type 4 — Event

Events happen over time.

Example

Meeting

↓

Alice approved Project Phoenix

↓

July 25

Events create history.

3. Why Graph instead of Tables?

Suppose CEO asks

"Which projects are affected if Legal Approval is delayed?"

With SQL

Need many JOINS.

Very slow.

Graph

Legal Approval

↓

Project Phoenix

↓

Deployment

↓

Engineering Team

Follow edges.

Answer immediately.

This is why graphs are useful.

4. What is a Node?

Everything important.

(Person)

Alice

(Project)

Phoenix

(Task)

Legal Approval

Each becomes

◦ Alice

◦ Phoenix

◦ Legal Approval

These circles are nodes.

5. What is an Edge?

An edge is simply a connection.

Alice

——works_on——>

Project Phoenix

Another

Project Phoenix

——depends_on——>

Legal Approval

Edges have labels.

Without labels,

AI cannot understand the meaning.

6. Why Extractors?

This is the biggest question.

Suppose you have

```
Alice approved Project Phoenix.
```

Can PostgreSQL understand that?

No.

Can pgvector understand that?

No.

Can a graph understand that?

No.

Someone must convert English into structured data.

That's exactly what the extractor does.

Input

```
Alice approved Project Phoenix.
```

Output

```
Entity
```

```
Alice
```

```
Entity
```

```
Project Phoenix
```

```
Relationship
```

```
Alice
```

```
approved
```

```
Project Phoenix
```

Now graph creation becomes possible.

7. Why Plugins?

Imagine every document.

Meeting

Decision

Owner

Action Item

Contract

Customer

Amount

Expiry

Financial

Revenue

Profit

Budget

Engineering

Repository

PR

Deployment

Release

One extractor cannot understand everything.

So Company Brain loads different plugins.

Meeting Plugin

Financial Plugin

Contract Plugin

Engineering Plugin

HR Plugin

Each specializes in one document type.

8. Why Entity Resolution?

Suppose company has

Microsoft

Another document

MSFT

Another

Microsoft Corporation

Without Entity Resolution

3 companies

Graph becomes wrong.

Entity Resolver merges

↓

Microsoft

One node.

9. Why Provenance?

Imagine AI says

Alice approved Project Phoenix.

CEO asks

"Where did you get that?"

Company Brain should answer

Meeting.pdf

Page 4

Paragraph 2

Without provenance

Nobody trusts AI.

Every node.

Every edge.

Every fact.

Must remember

Source Document

Page

Connector

Timestamp

10. Why Confidence?

Suppose OCR reads

\$2.8M

Maybe confidence

99%

Another scanned document

Confidence

54%

Should both be treated equally?

No.

Store

Confidence = 0.99

Confidence affects

Reasoning

Search

Recommendations

11. Why Versioning?

Yesterday

Budget

\$2.8M

Today

Budget

\$3.5M

Should old value disappear?

No.

Store

Version 1

↓

\$2.8M

Version 2

↓

\$3.5M

Now Company Brain knows history.

12. Why Incremental Sync?

Imagine

100 million documents.

One document changes.

Should AI process

100 million again?

No.

Knowledge Sync detects

FILE_UPDATED

Only

Budget.xlsx

runs through pipeline again.

Huge performance improvement.

13. Why Reasoning?

Suppose graph

Project Phoenix

↓

depends_on

↓

Legal Approval

Legal Approval



Overdue

Nobody wrote

Project Phoenix



At Risk

Reasoner infers



New knowledge

This is intelligence.

14. Why Impact Analysis?

Suppose policy changes.

Instead of searching

100,000 documents,

Graph walks

Policy



Engineering



VPN



IT



Employees

Instantly knows impact.

15. Why Organizational Health?

CEO doesn't care about

1 million nodes.

CEO wants

Knowledge Quality

87

and

Top Problems

12 Missing Owners

8 Missing Policies

15 Outdated Documents

Health converts graph into business metrics.

16. Why Decision Intelligence?

Suppose six months later

CEO asks

Why did we migrate to AWS?

Decision Intelligence stores

Decision

Reason

Evidence

Meeting

Owner

Expected Outcome

Institutional memory.

17. Why Recommendations?

Graph says

Project

↓

No Owner

Recommendation

Assign Owner

Graph says

Contract expires

10 days

Recommendation

Review Contract

Knowledge becomes actionable.

18. The Most Important Concept: Company Memory

This is the idea that differentiates your product.

Traditional RAG works like this:

Question

↓

Search

↓

Answer

↓

Forget

Every new question repeats the process.

Company Brain works like this:

Document

↓

Extract

↓

Understand

↓

Build Knowledge Graph

↓

Remember Forever

↓

Reason

↓

Improve

↓

Use Again

The company becomes smarter over time because it **remembers** what it has learned.

19. The Complete Flow

Google Drive / Slack / Jira / Email



Module 1

Extract text from files and messages



Module 2

Create chunks, embeddings, metadata



Retrieve relevant content



Module 3

Hybrid Extractors



Extract Entities



Extract Relationships



Extract Facts



Extract Events



Entity Resolution



Create Graph Nodes



Create Graph Edges



Store Provenance



Detect Conflicts



Reason Over the Graph



Analyze Business Impact



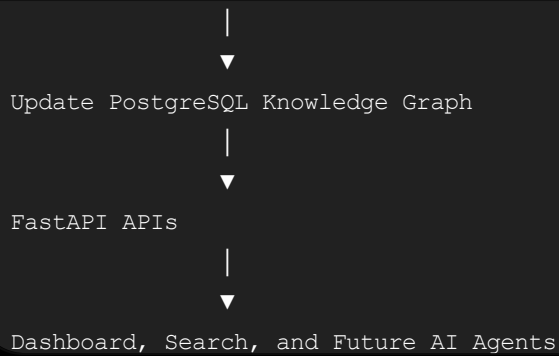
Measure Organizational Health



Store Decisions



Generate Recommendations



Two advanced concepts you should add to your understanding

These are not separate modules, but they are essential design principles.

A. Provenance is attached to everything

Every entity, relationship, and fact should be traceable back to its source.

For example:

```
Entity:
Project Phoenix

Created From:
Meeting.pdf

Connector:
Google Drive

Page:
3

Extracted At:
2026-07-25 13:45 UTC

Confidence:
0.97
```

This is what allows Company Brain to answer "**How do you know this?**"

B. The graph is a living model

The graph is **not built once**.

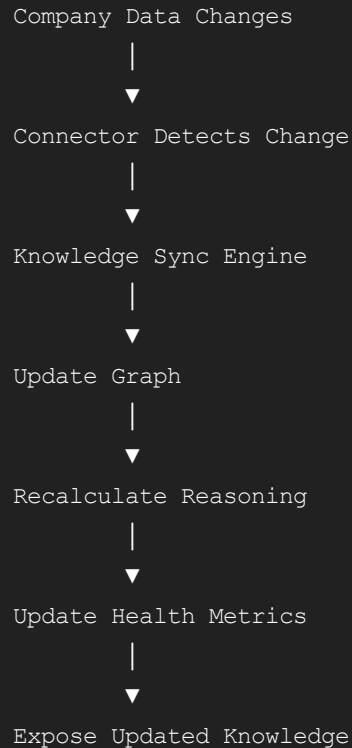
Every time a file is:

- Created
- Updated

- Deleted
- Permissions changed

the Knowledge Sync Engine updates only the affected entities, relationships, and facts.

So the lifecycle is:



This is what makes Company Brain a **living organizational memory** rather than a static database or a simple document search system.